



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Experiment No. 7

Aim: Perform DOM Traversal and Updates; Develop a To-Do List App with Add, Edit, and Delete Features

Software / Tools Required:

1. Visual Studio Code
2. Google Chrome
3. HTML5
4. JavaScript (ES6)

Theory: DOM (Document Object Model) represents an HTML document as a tree of elements. JavaScript can access and modify these elements dynamically.

a) DOM Traversal:

1. `getElementById()` is used to access HTML elements.
2. `querySelectorAll()` is used to select multiple elements.

b) DOM Manipulation:

1. `createElement()` is used to create new elements.
2. `appendChild()` is used to add elements.
3. `remove()` is used to delete elements.
4. `innerText` is used to display or modify text.

c) Event Handling:

1. `addEventListener()` is used to handle user actions.
2. The click event is used to perform operations when a button is clicked.

7.1 Implementation : To-Do List

A simple To-Do List application was developed using HTML, CSS and JavaScript. It allows the user to add, edit and delete tasks.

The following functions were used:

1. `getElementById()` – to access HTML elements.
2. `addEventListener()` – to handle button clicks.
3. `createElement()` – to create new list items and buttons.
4. `appendChild()` – to add tasks to the list.
5. `remove()` – to delete tasks.
6. `innerText` – to display and edit task names.



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7.2 Case Study : Registration Form

A Registration Form was created using HTML, CSS and JavaScript. It contains fields such as Firstname, Lastname, Birthday, Username, E-mail, Website, Password and Re-password along with a terms and conditions checkbox.

JavaScript was used to:

1. Access form elements using DOM methods.
2. Validate required fields and email.
3. Check password and re-password.
4. Handle **change** and **submit** events.
5. Display error messages for invalid input.

Conclusion: The practical was completed successfully. DOM traversal and manipulation were implemented using JavaScript. A To-Do List was developed with Add, Edit and Delete features, and a Registration Form was created with user input validation and event handling.



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4/5 PAGES OF SCREEN ACCESS. PROCEED TO NEXT

1 USER REGISTRATION

FORMAL USER'S REGISTRATION FORM ENTRY

1/ FULL NAME (20-100)

2/ USERNAME (10-100)

3/ DATE OF BIRTH (10-100)

4/ EMAIL ADDRESS (100-100)

5/ ACCESS CODE / PASSWORD (10-100)

ADD NEW INDICATES COMPULSORY FORM FIELDS

INITIALIZE USER PROGRAM

TASKS.EXE // V1.0.0

TASKS.EXE

ENTER A NEW TASK

ADD TASK

NO PENDING TASKS

ADD your TASKS here please to get started

ADD: 00:00:00

STATUS: 00:00:00



संयुक्त विश्वविद्यालय

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